UNIT 4 · 9 hours

n-Channel Enhancement MOSFET: Working and Characteristics



Transistors

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

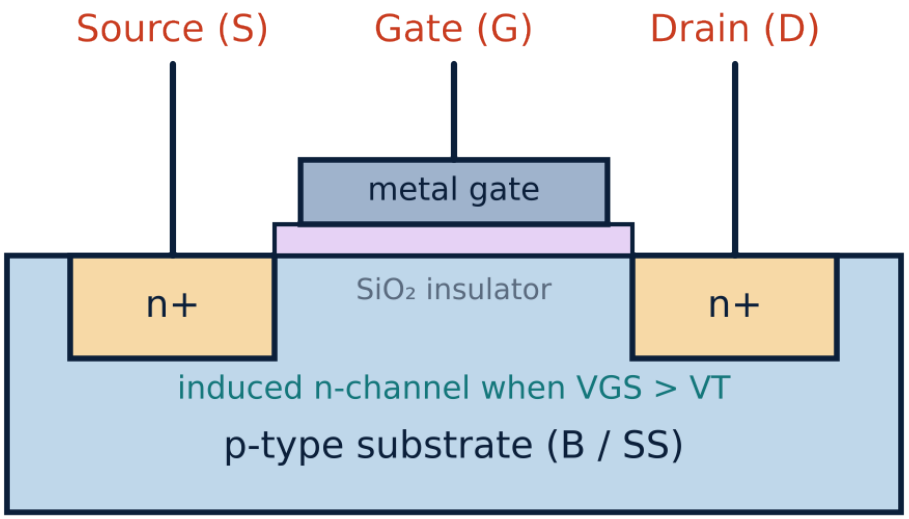
- Describe the structure of the n-channel enhancement MOSFET
- Explain channel formation and the threshold voltage
- Explain pinch-off and saturation
- Interpret drain and transfer characteristics

Syllabus context

Construction and operation, BJT as a switch, working and characteristics of n-channel enhancement type MOSFET, common source amplifier, DC load line analysis, MOS inverter.

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Structure



- p-type substrate (body) with two n+ regions: source and drain.
- Gate: metal over a thin SiO_2 insulator: no gate current.
- Four terminals: G, D, S and substrate (SS).
- No channel exists at VGS = 0: 'enhancement' type.

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Operation

1

VGS = 0

No channel; the two n+ regions are separated by the p-substrate: ID = 0.

2

VGS > VT

Positive gate repels holes and attracts electrons to the surface: an n-channel forms.

3

More VGS

More carriers; channel conductivity and ID rise.

4

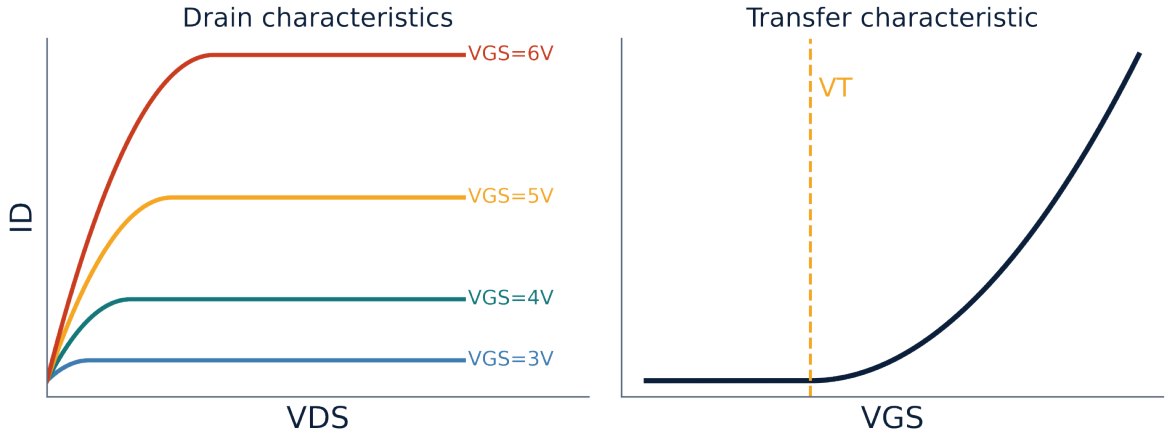
VDS rises

Channel narrows near the drain (pinch-off); ID saturates.

Pinch-off at VDS(sat) = VGS − VT; typical VT ≈ 1–3 V.

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Drain and transfer characteristics



- Cut-off: VGS < VT, ID = 0.
- Ohmic (triode): VDS < VGS − VT: ID = k[2(VGS − VT)VDS − VDS²].
- Saturation: VDS ≥ VGS − VT: ID = k (VGS − VT)².
- k ≈ 0.278 mA/V² in the notes' example.

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Regions of operation

Region	Condition	ID	Use
Cut-off	VGS < VT	0	Switch OFF
Triode / ohmic	VGS > VT, VDS < VGS − VT	$k[2(VGS-VT)VDS - VDS^2]$	Switch ON, resistor
Saturation	VGS > VT, VDS ≥ VGS − VT	$k(VGS - VT)^2$	Amplifier

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Numerical: drain current

Problem n-channel enhancement MOSFET, $k = 0.278\ mA/V^2$, $VT = 2\ V$, $VGS = 6\ V$. Check the region and find ID if $VDS = 8\ V$.

- $VGS - VT = 4\ V$; $VDS = 8\ V \geq 4\ V \rightarrow$ saturation
- $ID = k\,(VGS - VT)^2 = 0.278 \times 4^2$

Answer: ID = 4.45 mA

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MOSFET versus BJT

MOSFET

- Voltage controlled (VGS)
- Very high input resistance
- Unipolar (one carrier)
- Small area; dominant in ICs

BJT

- Current controlled (IB)
- Moderate input resistance
- Bipolar (two carriers)
- Higher gm for same current

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Key takeaways

- Gate insulated by SiO_2: no gate current
- Channel forms for VGS > VT
- Saturation: $ID = k(VGS - VT)^2$, $VDS(sat) = VGS - VT$
- Voltage-controlled device with very high input impedance

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Practice questions

Q1. Explain channel formation in an n-channel enhancement MOSFET.

Q2. $k = 0.5\ mA/V^2$, $V_T = 1.5\ V$, $V_{GS} = 4\ V$ in saturation. Find I_D.

Q3. Define threshold voltage and pinch-off.

Q4. Why is it called enhancement type?